

Determining the Crystalline Structure of Metals through X-Ray Diffraction

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Abstract

This experiment set out to measure X-ray diffraction patterns of various solid samples in order to describe their crystal structure at the atomic/molecular level. An already calibrated Siemen !!!!!MODEL NAME!!!! X-Ray diffractometer was used to send X-rays into Copper (Cu), Nickel (Ni), CuNi alloy, Titanium (Sn), Lead (Pb), PbSn alloy and two mystery samples named A and B.

Any crystalline solid consists of a regular 3-dimensional packing of cells, the shape of which corresponds to one of fourteen Bravais lattices. These 3D patterns create multiple parallel planar layers within the crystal of various orientations. X-Rays travelling through these planes will diffract off the atoms and constructively interfere at the detector, creating a peak in photon count in the observed diffraction pattern. From that peak, Bragg's law $n\lambda = 2d \sin \theta$ (where $n = 1$ for a first-order plane) can be used to deduce the inter-planar distance d of the diffraction plane of the crystal. Using this inter-planar distance and the corresponding Miller indices, the Bravais lattice can be determined and the lattice parameter(s) calculated. The Cu and Ni samples were found to have a Bravais lattice of [...], with parameters [...] respectively. The alloy sample of these two was found to have [...]. The Sn and Pb samples were found to have a lattice of [...] and parameters [...] respectively, and their alloy [...], [...]. Finally, the mystery samples A and B were determined to be [...] and [...], with lattices [...] and parameters [...], respectively.

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1 Introduction

The discovery of X-rays led to W.H. and W.L. Bragg researching the inner workings of X-ray diffraction. It was found a solid can serve as a diffraction grating for x-rays of similar lengths to the inter-atomic distances of the solid. Similarly to how light diffracts, one can obtain a relation between the inter-atomic distance d , the incident angle θ , and the wavelength λ :

$$n\lambda = 2d \sin(\theta) \quad (1)$$

with $n = 1$ for our purposes. Equation 1 is formally called Bragg's Law. Its utility becomes apparent when one ponders the crystalline structure of solids. Suppose the individual atoms of a crystal could be observed. While varying the angle the observer makes with the plane of the solid, there will be an angle where the atoms appear to line up. Given that a crystal is 3-dimension, the atoms will be aligned in parallel planes. Not all observed planes are equal however. Figure 2 shows a plane formed from a cubic arrangement of atoms. Notice that a plane could also have been formed from a face of the cube (in the y-z plane for example), or by aligning four diagonal atoms (the plane could be cutting through the y-axis and x-axis, for example). These planes are differentiated by their 'Miller index'. In Figure 2, [!!!EXPLAIN HOW TO GET MILLER INDEX!!!] Crystal are made of cells which can borrow one of fourteen 'Bravais lattices', each grouped into one of seven crystal systems. Figure 1 shows these lattices and their groups. The Miller indices can be related to the inter-planar distance d , in equation 1, by various equations depending on the Bravais lattice. In a cubic crystal system for example, the relation is given by

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2} \quad (2)$$

where h, k, l are the miller indices and a is the len of the cube edge. Similar equations for different crystal systems are given in Appendix ??.

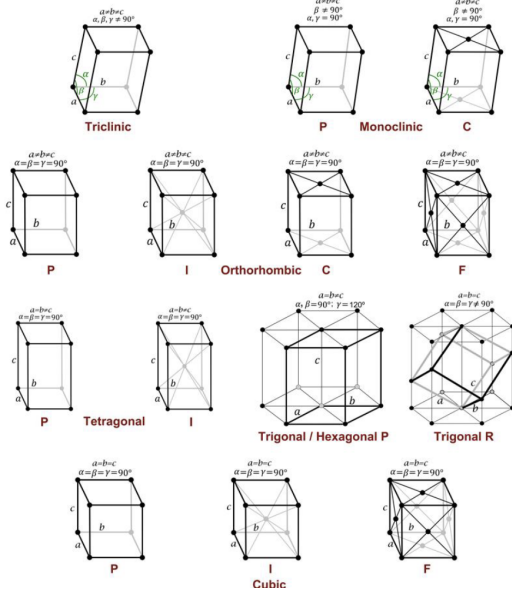


Figure 1: The fourteen Bravais lattices with their seven associated groups.

2 Methods

A pre-calibrated Siemens diffractometer was used to produce X-Rays, send them at the samples, and detect the diffractions. The diffractometer consisted primarily of an X-ray tube, a sample mounting plate, and a photon detector. The X-ray tube used Copper K_α radiation and was cooled using water. Two sets of filters of 0.6 mm were present on the X-ray tube and the detector, to prevent stray rays from leaving the tube and entering the detector. During data acquisition, the diffractometer was set to 40 kV and 40 mA. The mounting plate kept the sample in place with a spring-loaded mechanism. The samples were centered by aligning the middle of the sample with a red line marked on the mounting plate. The diffractometer changes the θ by rotating the mounting plate by some $\Delta\theta$ and moving the detector by $2\Delta\theta$. A computer was connected to the diffractometer, with which a .dql measuring parameter file was given to the diffractometer and data was obtained from the detector as .RAW files. The measuring parameters were an initial angle of $2\theta_0 = 5.00^\circ$, angle increments of $\Delta\theta = 0.05^\circ$, with 2.00 s at each step, with a final angle of $2\theta_f = 110.00^\circ$. The total measuring time is roughly 1 h 05 m using the aforementioned parameters. Data acquisition was simple: Mount the desired sample onto the mounting plate, centering it with the red line. Then, on the

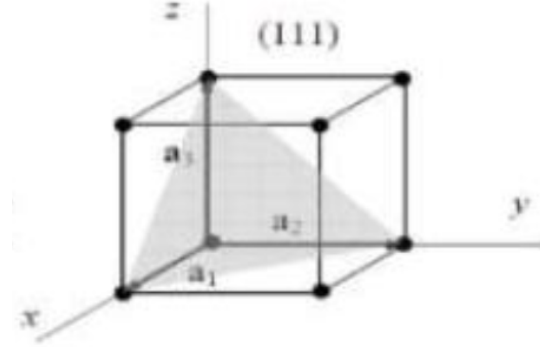


Figure 2: Plane corresponding to Miller index 111.

computer, open the software program [!!!NAME!!!], select the desired parameter file, name the trial, and execute the job. After the job finishes, convert the .RAW file to .UXD. Finally, the .UXD files were transferred from the lab computer onto the experimentalists's for the data analysis. Three trials were done for Ni and Cu samples, though the data for the first Cu trial did not save correctly. Two trials were done for Pb, Sn, A and B. For each of the alloys (including different percentage of mixing) one trial was done. It was determined that the uncertainty due to random errors would be less than that due to noise (could expand on this).

3 Results

From the .UXD files of a sample, the plot of light intensity against θ can be done. Figure ?? show this plot for the Copper and Nickel samples. Using prominence 50, local maxima of light intensities can be located and indexed to find angles where diffraction in planes occur. Every peak in the plot is then fitted to a gaussian curve, centered at a local maxima, and every curve is summed to obtain a plot of summed gaussians of intensity against 2θ . Figure ?? shows the summed gaussians for Copper and Nickel. The mean angle for every peak as well as its uncertainty can then be obtained from the corresponding gaussian. Table

4 Discussion

5 Conclusion